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MyDesign Portfolio

Element L Initial Template

Element L: Presentation of Designer's Recommendations

Content of Recommendations (L1, L2)

Earlier on (in K3, if you completed it) you provided general lessons learned related to engineering design, and now (in L2) you are asked to provide recommendations for continuing work on this project in particular.

Answer as many of these questions as you can with specific recommendations based on your testing data, expert/stakeholder feedback, and any new research you've gathered. These recommendations may include process improvements (how you worked) and/or project improvements (actual design specific decisions you made). Provide recommendations to someone doing a similar project that might build or continue upon your work. For example: "Taking into account <something we did>, we believe doing _____ would improve _____ because _____."

- What could be done next to go further?
 - To go further, you could conduct more tests in a body of water to collect more data on how effective the ROV is at collecting microplastics. You could then make more iterations to your ROV as you observed what was effective and what wasn't, to hopefully maximize the amount of microplastics you can collect.
 - To put this into action, you can analyze the data you collected and find strong and weak points. If you didn't collect many microplastics, you could improve your collection device or try another iteration. If your ROV frame took on a lot of water, you could focus on tightening the joints. These changes, followed by more testing, could lead to your design improving greatly, and hopefully maximizing the effectiveness of your ROV
 - We learned from Josiah's advice on Element C, that we could have considered a way to prevent marine life from getting caught in our net. Perhaps we could add a light and sound effect to ward off at least

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some sea life, such as smaller fish or crabs. This could also eliminate the danger of larger sea life interfering with the ROV (but it could also make them want to investigate it).

- What would you keep if you did the project again?
 - Taking into account our collection device, we believe keeping our net orientation the same is the right decision because it proved to have no issues and made the ROV easy to maneuver.
 - Taking into account our frame, we believe keeping its design the same is the right decision because it was durable and evenly weighted.
- What would you change?
 - Making sure the ROV is balanced and taking into account the weight of the cable during the tests.
(To accomplish this we would do further water testing and experiment with our number of pool noodles and their placement).
 - The motors should be as close to where the water will be (the bottom) as possible to be able to move the ROV if it won't sink
(To accomplish this we would flip the depth motor and adjust the net tension so that it can't get caught in the motor).
 - Have it be more aerodynamic in the water because movement in the water proved slow.
(To accomplish this we would add smooth exterior parts to encourage water to flow around the ROV, making it better at cutting through the water).
 - A bigger plankton net might have improved our collection process.
(To accomplish this we would swap out the old net for the bigger net, before securing it in a way which would not get tangled in the motors).
- What would have made this project easier for you to complete?
 - Having an unlimited budget. If we had no budget, we could have had more materials to experiment with, and also could have had the best nets and fastest motors.
 - More testing time. If we had more time at Fort Mason, we could conduct more trials and have a larger data pool.



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- More access to expert feedback. If we had a ton of field experts giving us feedback whenever we wanted, we probably would have had a more refined final product or had to make fewer mistakes to get to the product we have now.

In order to earn 5's on Element L, you will need specific and detailed ways the project can be improved and a rationale for why they are needed.

Detailed Plans for Implementation of Improvements (L2)

Now that you have generated a list of specific recommendations with their rationale, go back under each and generate a consistently detailed list about HOW these details will be implemented. Refer back to Element I where you concluded with some of these ideas.